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3.1 Introduction

This report reflects my learning journey throughout the BIT1123 Object-Oriented Programming (Java) course. The primary objective of this assignment is to consolidate all practical tutorial exercises completed from Week 1 to Week 10 into a structured GitHub repository, demonstrating progress in Java programming, documentation, and version control management.

3.2 Knowledge Gained from Each Tutorial

Week 1: Set up Java Development Environment (JDK) and mastered basic syntax and control structures.

Week 2: Learned the core OOP concepts of classes, objects, state, and behaviors.

Week 3 & 4: Implemented custom constructors, method overloading, and object initialization.

Week 5: Applied encapsulation using access modifiers (private, public) and created proper getters and setters.

Week 6: Understood inheritance principles and built class hierarchies using derived classes (extends).

Week 7: Implemented dynamic polymorphism and method overriding to achieve runtime behavior modifications.

Week 8 & 9: Practiced modular code design using abstract classes and interfaces.

Week 10: Handled runtime errors effectively using try-catch blocks and exception handling mechanisms.

3.3 Challenges Encountered

Managing version control commands and resolving file path conflicts on GitHub.
Grasping the distinction between abstract classes and interfaces during implementation.
Structuring OOP code cleanly without over-complicating class relationships.

3.4 How the Challenges Were Overcome

Reviewed official Java and Git documentation to fix versioning errors.
Refactored code iteratively and debugged runtime errors using IDE debugging tools.

Consulted lecture notes and completed practical exercises to solidify core OOP concepts.

3.5 Improvements in Java Programming Skills

My coding efficiency and problem-solving abilities improved significantly. I transitioned from writing basic procedural scripts to designing modular, well-structured object-oriented programs.

3.6 Understanding of Object-Oriented Programming Concepts

I have developed a solid understanding of the four pillars of OOP:

1. Encapsulation: Protecting data through access control.
2. Abstraction: Hiding complex implementation details behind simple interfaces.
3. Inheritance: Promoting code reusability across parent-child classes.
4. Polymorphism: Allowing objects to take on multiple forms dynamically.

3.7 Future Learning Plans

In the future, I plan to explore advanced Java Data Structures, graphical user interface (GUI) frameworks, and backend application development using the Spring Boot framework.

3.8 Conclusion

This assignment provided valuable practical experience in both Java programming and software project organization. Building a complete portfolio on GitHub has prepared me for standard software development practices.

3.9 GitHub Repository URL

<https://github.com/Sharmin3679/Java-tutorial/blob/main/README.md>